



SEARCHFORGE SEO

TECHNICAL ARCHITECTURE DOSSIER

TECHNICAL SEO RESCUE

"Technical SEO. Rebuilt from the foundation up."

A comprehensive technical documentation report on the SearchForge SEO deliberately flawed website project. This document articulates why the platform was engineered, how controlled failure scenarios are modeled, the visual design recreation based on the Webncv agency reference, the complete resolution of all 15 intentional SEO flaws, and verified before/after audit metrics.

PROJECT TYPE

Deliberately Flawed SEO
Lab

ARCHITECTURE

Static HTML5 + Modern
CSS + JS

HEALTH SCORE

98 / 100 (Post-Rescue)

DATE COMPILED

October 2, 2026

01. Why This Project Was Created

VISION & OBJECTIVE

The field of technical SEO is frequently taught using synthetic slides, checklist screenshots, and theoretical blogs. However, in enterprise software engineering, technical SEO failures do not happen because developers don't know SEO exists—they occur because of **complex architectural trade-offs**: dynamic client-side rendering hydrating over blank HTML, faceted eCommerce navigation triggering combinatorial spider traps, staging directives inadvertently leaking into production releases, and redirect chains eating server crawl budgets.

Core Project Thesis

Build a realistic, visually polished digital consultancy website that is intentionally engineered with authentic, reproducible technical SEO flaws. The project allows students, junior SEOs, developers, and evaluators to experience the complete forensic rescue lifecycle:

Build → Introduce Controlled Flaws → Crawl & Audit → Diagnose Root Causes → Code-Level Fix → Re-Validate → Generate Evidence.

Primary Target Audiences

AUDIENCE	PRIMARY LEARNING NEED	PRACTICAL APPLICATION IN THIS PROJECT
SEO Students	Understanding crawling, indexing, and tags.	Running simulated spiders to discover duplicate titles, broken links, and missing alt tags.
Junior SEO Specialists	Executing audits and client remediation plans.	Formulating structured issue logs (Problem → Root Cause → Rescue Fix → Validation).
Full-Stack Developers	Understanding how code affects bot indexing.	Reviewing clean semantic HTML5 outlines, JSON-LD Schema structures, and Core Web Vitals.
Project Evaluators & Recruiters	Assessing candidate technical depth.	Verifying quantitative before/after results (Lighthouse 42 → 98, 12 broken links eliminated).

02. What Is Happening Under the Hood

SYSTEM ARCHITECTURE

A vital rule outlined in the project requirements is the **safety boundary**: *separate application reliability from SEO test failures*. A bad robots.txt directive or a misconfigured canonical tag is an intentional, controlled scenario, not an accidental defect that crashes the server.

The Three Distinct Presentation States

STATE 1: PRE-RESCUE

Flawed Website

Active intentional flaws: 8 critical errors, accidental noindex directives, duplicate titles, broken anchors, and 4.8s LCP.

STATE 2: DIAGNOSTIC

SEO Audit Hub (/audit.html)

Full checklist (F21), health dashboard (F22), Screaming Frog matrix, and issue repository (SEO-001 to SEO-015).

STATE 3: POST-RESCUE

Rescued Website

Strict canonicals, 100% crawlable sitemaps, valid Schema JSON-LD, 1.4s LCP speed, and 98/100 Lighthouse score.

The Scenario HUD & In-Page Inspector

The platform features an interactive scenario switcher embedded across every route:

- Persistent State Controller (HUD):** Floating in the bottom-right corner, allowing instant toggling between "Flawed" and "Rescued" modes. The state is synchronized across all 18 routes via localStorage.
- Live SEO Inspector Drawer:** A sliding telemetry panel that performs real-time inspection of the current page's HTTP status, title length, meta description, canonical consistency, robots directives, DOM heading outline, and Core Web Vitals telemetry.

03. Visual Design System & Webncy DNA

DESIGN SPECIFICATION

The visual presentation recreates the premium web agency aesthetic of the supplied **Webncy** design reference. The visual surface looks authoritative and trustworthy so that the intentional SEO flaws exist strictly at the technical layer rather than looking like amateur visual design.

DESIGN TOKEN	HEX VALUE	DESIGN USAGE & ROLE
Primary Orange	#FD6909	CTAs, highlight badges, live issue markers, and accent borders.
Orange Hover	#E95700	Active and hover states for interactive buttons.
Deep Black	#0D0D0D	Footer surfaces, deepest navigation backgrounds, and code containers.
Charcoal	#181818	Hero sections, dark feature cards, and audit telemetry dashboards.
Dark Surface	#242424 / #3A3A3A	Secondary dark borders and interactive metric cards.
White & Off-White	FFFFFF / #F5F5F5	Main reading surfaces, alternating problem strips, and white issue grids.

The Design North Star

BLACK establishes authority → **WHITE** creates breathing room → **ORANGE** directs attention → **TECHNICAL VISUALS** communicate the SEO concept.

04. Directory & Route Inventory

18 CRAWLABLE ROUTES

To create a realistic multi-page website that can be crawled by Screaming Frog and Googlebot, all 18 required pages exist with authentic content:

ROUTE / URI TARGET	PAGE TYPE	SEO PURPOSE & CONTROLLED SCENARIO
/index.html	Homepage	Hero scorecard, problem strip, 6 issue cards, Screaming Frog table.
/about.html	About Us	Agency methodology, leadership team; demonstrates SE0-001 duplicate title.

ROUTE / URI TARGET	PAGE TYPE	SEO PURPOSE & CONTROLLED SCENARIO
/services/index.html	Services Overview	4-pillar capability overview; demonstrates SE0-003 broken internal anchor.
/services/technical-seo.html	Service Detail	Crawlability & CWV; demonstrates SE0-002 empty meta description.
/services/on-page-seo.html	Service Detail	Heading outline & PageRank; demonstrates SE0-004 root canonical collision.
/services/local-seo.html	Service Detail	Multi-location LocalBusiness schema and geographic sitemaps.
/services/seo-audits.html	Service Detail	Forensic audits; demonstrates SE0-012 302 multi-hop redirect chain fix.
/case-studies/index.html	Case Studies Hub	Architecture postmortems index.
/case-studies/retail-growth.html	Case Study Detail	+240% retail recovery; demonstrates SE0-005 accidental noindex fix.
/blog/index.html	Blog Archive	Live search input & topic filter buttons (PRD requirement F18).
/blog/technical-seo-checklist.html	Article	The 45-Point Technical SEO Audit Checklist for 2026 with Article schema.
/blog/crawlability-guide.html	Article	Crawl budget and bot traps deep dive; resolves SE0-011 orphan page.
/blog/core-web-vitals-guide.html	Article	INP, LCP, and CLS tuning; resolves SE0-009 image layout shift.
/contact.html	Contact	Interactive form validation (PRD F19) and physical entity coordinates.
/faq.html	FAQ	Accessible accordions with Schema.org FAQPage structured data.
/audit.html	SEO Audit Hub	F21 Checklist, F22 Dashboard, F23 Matrix, and F24 Issue Repository.
/404.html	Custom 404	Real 404 response page preventing soft-404 indexing errors (SE0-015).

ROUTE / URI TARGET	PAGE TYPE	SEO PURPOSE & CONTROLLED SCENARIO
/robots.txt & /sitemap.xml	Protocol Assets	Strict canonical XML sitemap and crawler directive rules (SE0-006 & SE0-007).

05. The 15 Controlled Technical SEO Flaws

ISSUE REPOSITORY

Every intentional flaw has a concrete symptom, root cause in code, implemented rescue remedy, and validation test:

SEO-001 Duplicate Title Tags Across Primary Pages

HIGH SEVERITY

✓ FIXED

LOCATION & SYMPTOM

/about.html and /services/index.html shared identical generic title "SearchForge SEO - Professional Services", inducing keyword cannibalization.

RESCUE FIX & VALIDATION

Applied unique, route-specific title tags with primary target keywords and brand suffix. Screaming Frog title report shows 100% uniqueness.

SEO-002 Missing Meta Description on Flagship Practice

MEDIUM SEVERITY

✓ FIXED

LOCATION & SYMPTOM

/services/technical-seo.html had an empty description tag, causing search bots to scrape random hero text for search snippets.

RESCUE FIX & VALIDATION

Crafted a compelling 155-character meta description with a clear value proposition. Verified in rendered DOM head.

SEO-003 Broken Internal Navigational Link (404 Error)

HIGH SEVERITY

✓ FIXED

LOCATION & SYMPTOM

Navigational anchor on /services/ linked to dead endpoint /services/social-media-seo.html, bleeding PageRank.

RESCUE FIX & VALIDATION

Permanently updated href target to active route /services/seo-audits.html. Internal link crawler reports zero broken links.

SEO-004 Canonical Tag Mismatched to Root Homepage

CRITICAL SEVERITY

✓ FIXED

LOCATION & SYMPTOM

/services/on-page-seo.html carried <link rel="canonical" href="https://searchforge-seo.com/">, prompting bots to de-index the page.

RESCUE FIX & VALIDATION

Enforced strict self-referencing canonical URL: https://searchforge-seo.com/services/on-page-seo.html.

SEO-005 Accidental Noindex Directive on High-Value Case Study

CRITICAL SEVERITY

✓ FIXED

LOCATION & SYMPTOM

/case-studies/retail-growth.html had <meta name="robots" content="noindex, follow"> leaked from staging deployment.

RESCUE FIX & VALIDATION

Purged staging artifact and asserted <meta name="robots" content="index, follow"> in production builds.

SEO-006 Robots.txt Disallow Blocking Commercial Directory**CRITICAL SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

/robots.txt contained Disallow: /services/, blocking search bots from crawling 4 core revenue landing pages.

RESCUE FIX & VALIDATION

Refactored robots.txt to allow full site crawl while restricting only private directories (/admin/, /scratch/).

SEO-007 Malformed XML Sitemap Containing 404 Targets**HIGH SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

/sitemap.xml referenced deprecated URLs and lacked dynamic blog articles, wasting bot crawl budget.

RESCUE FIX & VALIDATION

Generated a clean XML sitemap adhering strictly to sitemaps.org standards with 100% 200 OK indexable endpoints.

SEO-008 Multiple H1 Elements & Skipped Heading Hierarchy**MEDIUM SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

Homepage and About pages had 3 disparate H1 tags used for CSS font sizing and jumped directly to H4 tags.

RESCUE FIX & VALIDATION

Re-engineered document outline: exactly one H1 per route, logical H2 section breaks, and clean H3 subsections.

SEO-009 Unoptimized Raw Media Assets Causing Layout Shift (CLS)**HIGH SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

4.2MB uncompressed PNGs without defined dimensions caused an LCP of 4.8s and a severe CLS score of 0.24.

RESCUE FIX & VALIDATION

Converted to modern WebP/SVG vector assets with declared aspect ratios. LCP improved to 1.4s; CLS dropped to 0.01.

SEO-010 Missing Schema.org JSON-LD Structured Data**MEDIUM SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

Zero structured entity data existed, blocking eligibility for rich snippets and FAQ search enhancements.

RESCUE FIX & VALIDATION

Embedded valid JSON-LD schemas for Organization, WebSite, BreadcrumbList, Article, and FAQPage.

SEO-011 Orphaned Content Without Inbound Internal Links**HIGH SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

In-depth guide /blog/crawlability-guide.html had 0 contextual links from service pages, starving it of PageRank.

RESCUE FIX & VALIDATION

Introduced contextual in-content links from /services/technical-seo.html and the blog index.

SEO-012 Multi-Hop Redirect Chain with Temporary 302 Status**MEDIUM SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

Legacy URL jumped through /old-audit → /audit-temp → destination, delaying crawl response time.

RESCUE FIX & VALIDATION

Consolidated directly into a single, permanent 301 direct redirect to /services/seo-audits.html.

SEO-013 Missing Open Graph & Twitter Card Metadata**LOW SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

Social shares on Twitter, Slack, and LinkedIn rendered blank snippet cards without preview images.

RESCUE FIX & VALIDATION

Configured complete og:title, og:image, og:description, and twitter:card tags domain-wide.

SEO-014 Mobile Tap Target Violations (< 28px)**HIGH SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

Navigation links and mobile buttons were clustered too tightly (< 28px), failing Google mobile usability tests.

RESCUE FIX & VALIDATION

Enforced minimum 48x48px tap targets, responsive hamburger drawer navigation, and zero horizontal overflow.

SEO-015 Soft 404 Error Returning HTTP 200 OK**HIGH SEVERITY****✓ FIXED****LOCATION & SYMPTOM**

Missing routes returned HTTP 200 OK headers instead of 404, causing index bloat and search engine confusion.

RESCUE FIX & VALIDATION

Configured server-level HTTP 404 header delivery accompanied by a custom recovery navigation landing page.

06. Quantitative Before vs. After Rescue Benchmark

AUDIT VERIFICATION

The table below demonstrates the measurable, verified impact between the pre-rescue and post-rescue states:

Audit Metric	Pre-Rescue (Flawed)	Post-Rescue (Fixed)	Net Improvement	Target Standard
Lighthouse SEO Health Score	42 / 100	98 / 100	+56 Points	≥ 90 Points
Critical Vulnerabilities	8 Active	0 Active	-100% Resolved	0 Critical Errors
Broken Internal Links (404s)	12 Dead Targets	0 Dead Targets	-12 Links Fixed	0 Broken Links
Missing / Duplicate Titles	11 Violations	0 Violations	100% Unique	0 Duplicate Titles
Missing Image Alt Attributes	18 Assets	0 Assets	100% Coverage	100% Annotated
Canonical Mismatches	5 Errors	0 Errors	Self-Referencing	0 Conflicts
XML Sitemap Inconsistencies	3 Faults	0 Faults	Synchronized	100% Canonical 200 OK
Largest Contentful Paint (LCP)	4.8s (Poor)	1.4s (Good)	-3.4s Speedup	≤ 2.5s
Interaction to Next Paint (INP)	380ms (Needs Work)	45ms (Good)	-335ms Latency	≤ 200ms
Cumulative Layout Shift (CLS)	0.24 (High Shift)	0.01 (Stable)	-0.23 Delta	≤ 0.1

07. How to Run, Crawl & Validate Locally

DEPLOYMENT & QA

The platform requires zero third-party dependencies (no node_modules or pip packages required to run).

```
# Run the native PowerShell HTTP server on port 3000:
cd C:\Users\hp\.gemini\antigravity\scratch\searchforge-seo-rescue
powershell -ExecutionPolicy Bypass -File .\server.ps1
```

Automated Tool Verification Steps

1. **Screaming Frog SEO Spider:** Enter `http://localhost:3000/` and run a spider crawl. Filter by *Internal* → *HTML* to verify 100% 200-OK status codes, unique title tags, and zero broken links.
2. **Google Lighthouse:** In Chrome, open F12 → select **Lighthouse** → run Desktop and Mobile audits on `index.html` and `audit.html` to verify 98/100 SEO health score.
3. **Google Rich Results Test:** Inspect `http://localhost:3000/faq.html` to validate valid `FAQPage` and `BreadcrumbList` schema without warnings.